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Using device: cpu
Using single-amplitude feyn data from /var/folders/1s/69py7bdx0g7g0t4_8rjh6g9h0000gn/T/single_amplitude_eval_f2ywx53/first_amplitude.csv
Imported 1 token rows
Wrote raw expression data to /Users/kymani/Documents/New_Scattering_AI/Repo/gravity-evaluator-improvements/after/user_beam20.csv
Wrote normalized token data to /Users/kymani/Documents/New_Scattering_AI/Repo/gravity-evaluator-improvements/after/user_beam20_tok.csv
Input token limit: unlimited; longest input: 899 content / 901 with BOS/EOS
Numerical evaluator: gravity (processes={'3s2h': 1}; reference_modes=first,last,random; gauge_shift=True; tol_abs=2e-09; tol_rel=2e-08)
Loading model from /Users/kymani/Documents/Big_YM_Data_Sets/Cyclic_Gravity/Model/best_model.pt

Running decode mode: beam; max_length=256; beam_size=20; mask_invalid_tokens=False
wrote human CSV to /Users/kymani/Documents/New_Scattering_AI/Repo/gravity-evaluator-improvements/after/user_beam20_beam_human.csv

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[beam] n=1
top-1 equivalent      :    0 / 1  (0.0%)
any-beam equivalent:    0 / 1  (0.0%)
avg valid beams      : 20.0
shorter equivalent outputs: 0 / 1
copied inputs: 0 / 1
independent target: unavailable; exact-target metrics are not applicable

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Example predictions (1 rows):
row 0 [beam]
target simple :
input scram   : ((e_4 · e_5)*(e_4 · e_5))/6 + ((e_4 · p_5)*(e_4 · p_5)*(e_5 · p_4)*(e_5 · p_4))/(6*(p_4 · p_5)*(p_4 · p_5)) - ((e_4 · p_2)*(e_4 · p_2)*(e_5 · p_3)*(e_5 · p_3))/((p_2 · p_4)*(p_3 · p_5*p_4*p_5*p_4*p_5))
top1 pred     : (p_2·F_4·p_5*p_1·F_4·p_5*p_1·F_5·p_4*p_1·F_5·p_4)/(p_1·p_4*p_1·p_5*p_2·p_3*p_4·p_5*p_4·p_5*p_4·p_5)
top1 metrics  : token=None string=None num_simple=None num_scr=0
any-beam      : token=None string=None num_simple=None num_scr=0
wrote detail results to /Users/kymani/Documents/New_Scattering_AI/Repo/gravity-evaluator-improvements/after/user_beam20_beam_results.csv

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Wrote summary to /Users/kymani/Documents/New_Scattering_AI/Repo/gravity-evaluator-improvements/after/user_beam20_summary.csv
Total wall time: 9.32s

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Terminal output PDF: /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_testing/outputs/Gravity_5_Evaluation_Information_2026-09-23_01-22-54-101126.pdf

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